Coordinates and Straight-Line Graphs

Name _____ Date _____ No calculator Version 1

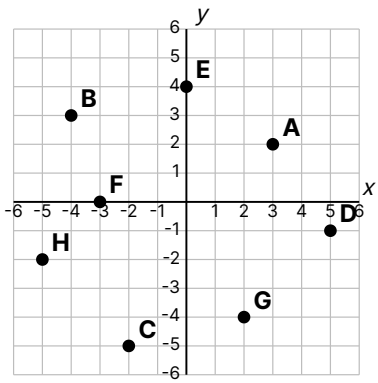
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KEY WORDS coordinates *(taunga)* · table of values *(tūtohi uara)* · gradient *(rōnaki)* · y-intercept *(pūtahi-y)*

A Coordinates ●○○

Remember: The first number is how far across (x) and the second is how far up or down (y). The point $(3, -2)$ is 3 right and 2 down.

1. Write the coordinates of each point.



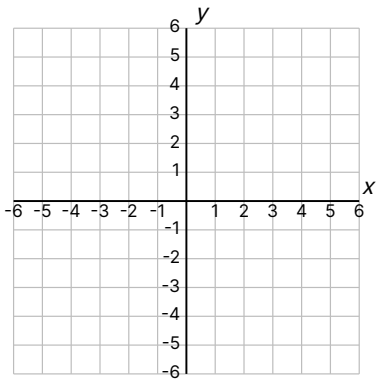
a) A _____ **b)** B _____

c) C _____ **d)** D _____

e) E _____ **f)** F _____

g) G _____ **h)** H _____

2. Plot these points on the grid and label each one with its letter: P $(4,\ 1)$, Q$(-3,\ 2)$, R$(-2,\ -4)$, S$(5,\ -3)$, T$(0,\ 3)$, U$(-5,\ 0)$.



B Tables of values ●●○

Remember: Put each value of x into the equation to find y.

3. Complete the table of values for each line.

a) $y = x + 2$

x	-2	-1	0	1	2
y					

b) $y = 2x$

x	-2	-1	0	1	2
y					

c) $y = 2x - 1$

x	-2	-1	0	1	2
y					

d) $y = -x + 3$

x	-2	-1	0	1	2
y					

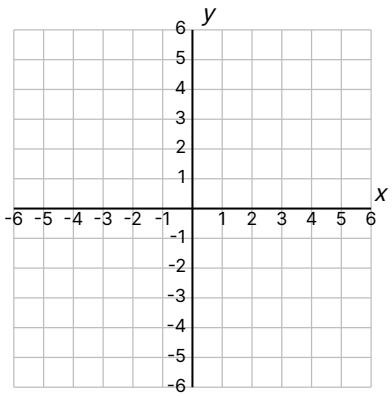
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C Drawing lines ●●●

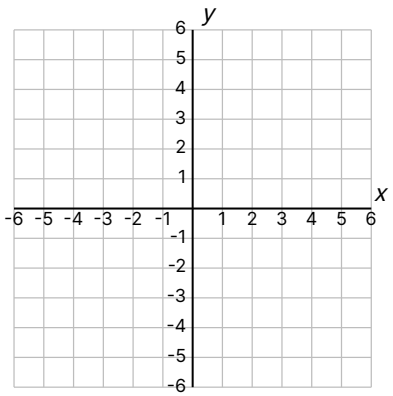
Remember: Make a table of values, plot the points, then rule a straight line through them.

4. Draw each line on its grid.

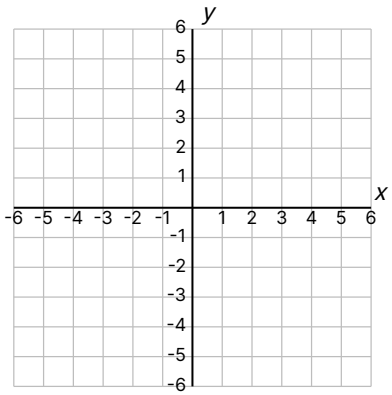
a) $y = x + 2$



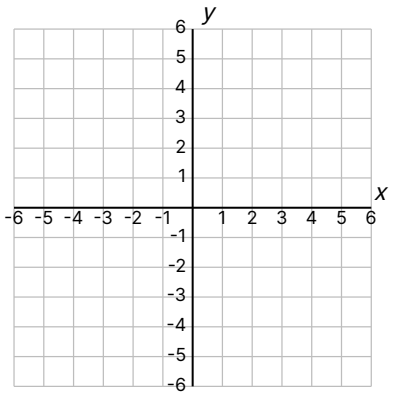
b) $y = 2x$



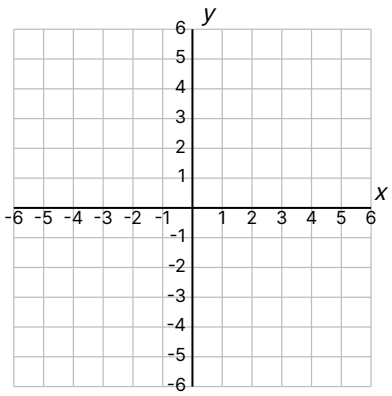
c) $y = 2x - 1$



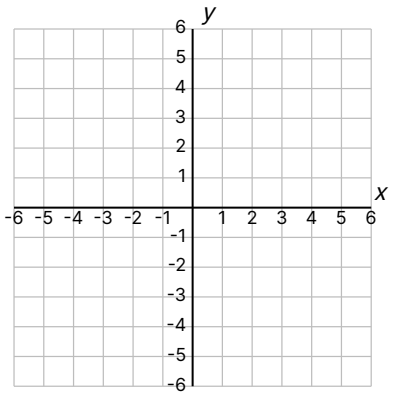
d) $y = -x + 3$



e) $y = x - 4$



f) $y = 3x - 2$



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D Gradient and y-intercept ●●○

> **Remember:** In $y = mx + c$, the gradient is m (how steep the line is) and the y-intercept is c (where it crosses the y-axis).

5. Write down the gradient and the y-intercept of each line.

a) $y = 2x + 5$ _____ **b)** $y = 4x - 1$ _____

c) $y = -3x + 2$ _____ **d)** $y = x - 6$ _____

e) $y = 5x$ _____ **f)** $y = \frac{1}{2}x + 3$ _____

g) $y = 7 - 2x$ _____ **h)** $y = -x$ _____

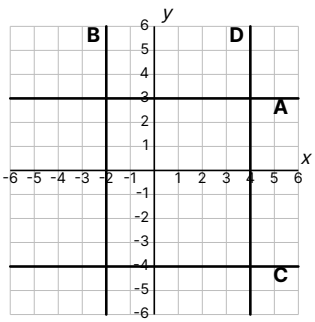
E Horizontal and vertical lines ●●○

> **Remember:** Every point on a horizontal line has the same y value. Every point on a vertical line has the same x value.

6. Write the equation of each line on the graph.

a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____



★ Extension

7. Does the point lie on the line? Show your working.

a) $(2,\ 7)$ and $y = 3x + 1$ **b)** $(4,\ 2)$ and $y = 2x - 5$ **c)** $(-1,\ 8)$ and $y = -2x + 6$

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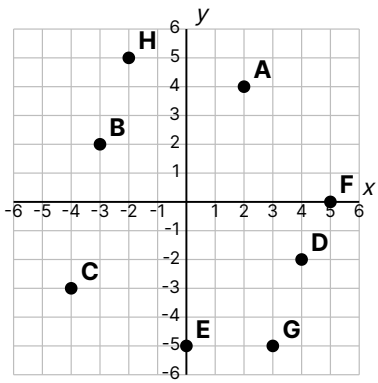
Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

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A Coordinates ●○○

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1. Write the coordinates of each point.



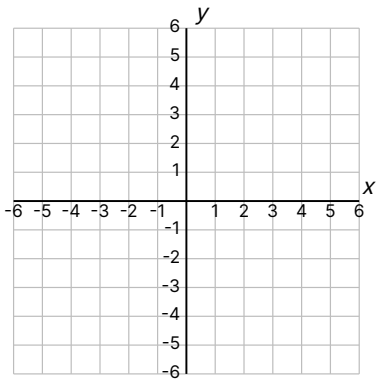
a) A _____ **b)** B _____

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2. Plot these points on the grid and label each one with its letter: P $(3,\ 2)$, Q$(-4,\ 1)$, R$(-3,\ -5)$, S$(2,\ -4)$, T$(0,\ -2)$, U$(4,\ 0)$.



B Tables of values ●●○

Remember: Put each value of x into the equation to find y.

3. Complete the table of values for each line.

a) $y = x + 3$

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y					

d) $y = -x + 4$

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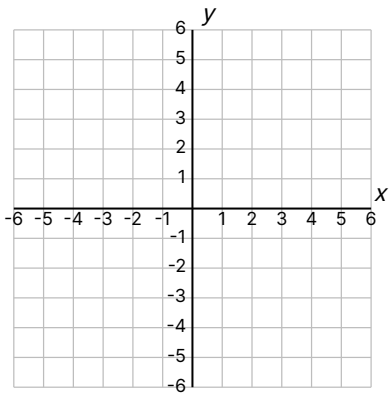
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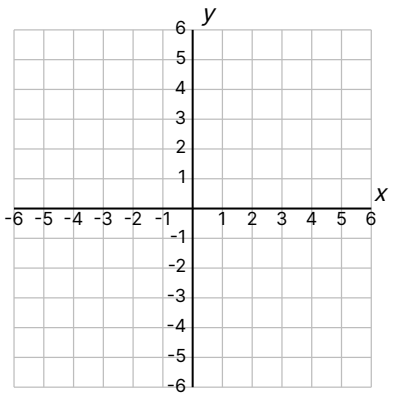
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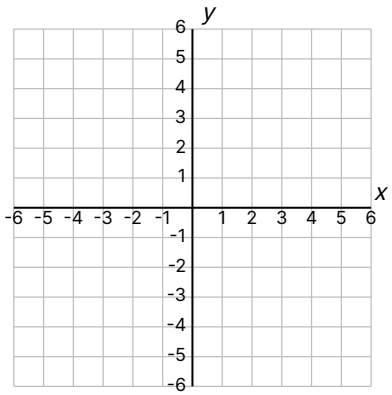
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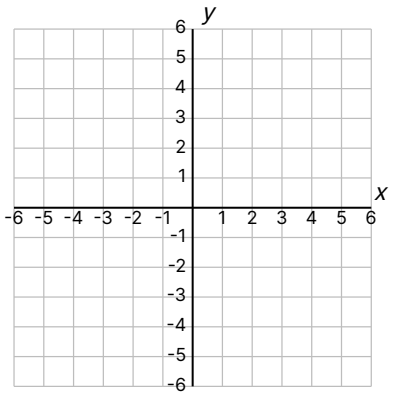
b) $y = 3x$



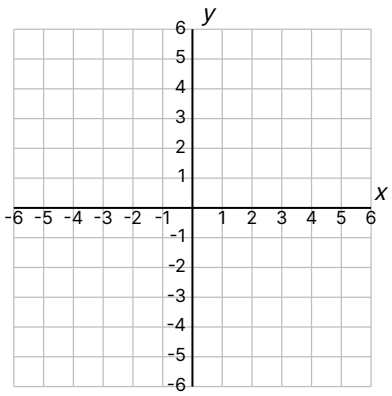
c) $y = 2x - 3$



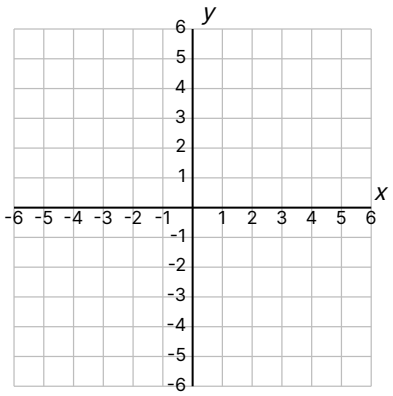
d) $y = -x + 4$



e) $y = x - 5$



f) $y = 3x - 1$



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5. Write down the gradient and the y-intercept of each line.

a) $y = 3x + 4$ _____ **b)** $y = 2x - 7$ _____

c) $y = -4x + 1$ _____ **d)** $y = x - 9$ _____

e) $y = 6x$ _____ **f)** $y = \frac{1}{3}x + 2$ _____

g) $y = 5 - 3x$ _____ **h)** $y = -x$ _____

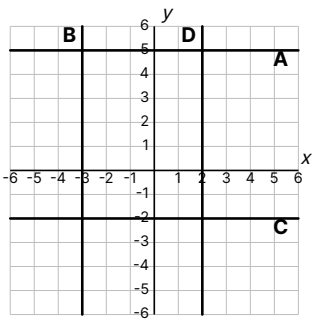
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a) Line A _____ **b)** Line B _____

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★ Extension

7. Does the point lie on the line? Show your working.

a) $(4,\ 11)$ and $y = 2x + 3$

b) $(3,\ 6)$ and $y = 3x - 4$

c) $(-2,\ 11)$ and $y = -3x + 5$

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